The paleomagnetism of Huronian red beds and Nipissing diabase; Post-Huronian Igneous Events and apparent polar path for the interval −2300 to −1500 Ma for Laurentia

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Thermal, chemical, and alternating field (and two-stage) cleaning treatments of Huronian sediments and Nipissing diabase (which intrudes the sediments) from the Cobalt area yield five directions of magnetizations (A–E) of high stability; A, B, C, and E are found in the sediments, and C, D, and E in the diabase. It is suggested that magnetization B (337°, + 52°; α_{95} = 8°; pole 158° E, 67° N) was acquired shortly after deposition of the Firstbrook beds (≃ −2288 Ma); magnetization C (259°, + 82°; α_{95} = 5°; pole 258° E, 42° N), found in both the diabase and sediments in contact with the diabase, was acquired during cooling following emplacement of the diabase (≃2150 Ma); and magnetizations D and E, yielding poles at 264° E, 15° S and 000°, 09° N respectively, were produced during the Hudsonian orogeny (−1850 to −1700 Ma). This interpretation resolves the previous inconsistencies between poles and age determinations. Good agreement between results from the Nipissing diabase and other igneous bodies indicate that widespread igneous events occurred in the time range approximately −2200 to −2100 Ma, immediately following deposition of Huronian sediments. This is referred to as 'Post-Huronian Igneous Events'. A proposed apparent polar path relative to Laurentia shows two distinct motions; for the 2300–1850 Ma interval, a latitudinal change (roughly along longitude 250° E) from high (≃70° N) to low (≃20° S) latitudes and, for the 1850–1500 Ma interval, a displacement along the present-day equator with first an eastward motion to about 000° longitude followed by a westward motion to 240° E longitude; the apex of the eastward excursion is given a date of ≃1750 Ma. It is possible that this reflects a rotation of Laurentia about a vertical axis at the time of and following the Hudsonian orogeny. Subsequent uplift and cooling would explain the many overprinted stable magnetizations yielding poles distributed along the equator (track 4). Latitude maps indicate that Laurentia was in high latitudes from 2200–2000 Ma and in intermediate to low latitudes from 1900–1500 Ma.

Des traitements thermiques, chimiques, et par champs alternatifs, en plus d'une méthode de désaimantation à deux étapes ont été appliqués à des sédiments Huroniens de la région de Cobalt, Ontario, ainsi qu'à des diabases du Nipissing qui coupent ces derniers. Cinq directions différentes d'aimantations (A–E) très stables ont ainsi été recouvrées; A, B, C, et E dans les sédiments et C, D, et E dans les diabases. Nous suggérons que l'aimantation B (337°, +52°, α_{95} = 8°, pôle 158° E, 67° N) fut acquise peu de temps après le dépôt des couches de Firstbrook (−2288 Ma); que l'aimantation C (259°, +82°, α_{95} = 5°, pôle 258° E, 42° N) que l'on retrouve dans les sédiments en contact avec les diabases et dans les diabases elles-mêmes fut acquise lors du refroidissement qui suivit l'emplacement de ces diabases, il y a 2150 Ma; et que les aimantations D et E donnant des pôles à 264° E, 15° S et 000°, 09° N respectivement retrouvent probablement leurs origines à l'orogénèse Hudsonienne.

Cette interprétation inédite met un terme au désaccord qui existait entre les données paléomagnétiques et radiométriques. Une excellente concordance entre les résultats des diabases du Nipissing et de plusieurs autres intrusions indique que des événements ignés eurent lieu dans l'intervalle de temps de −2200 à −2100 Ma, soit immédiatement après le dépôt des sédiments Huroniens. Nous référons alors aux 'Evénements ignées post-Huronien'. Nous proposons un tracé de la dérive apparente des pôles relatif à Laurentia qui révèle deux movements distincts; pour la période −2300 à −1850 Ma, un changement latitudinal (approximativement le long de la longitude 250° E) de hautes (70° N) à basses (20° S) latitudes et, pour la période −1850 à −1500 Ma, un déplacement le long du présent équateur d'abord vers l'est jusqu'à la longitude 000°, et ensuite vers l'ouest jusqu'à la longitude 240° E. La date du point de retour de l'excursion vers l'est est à peu près −1750 Ma. Ce déplacement le long de l'équateur représente peut-être une rotation de Laurentia sur elle-même, pendant et après l'orogénèse Hudsonienne. Le soulèvement et le refroidissement subséquents expliqueraient alors les surimpressions de plusieurs aimantations stables qui échelonnent des pôles le long du présent équateur. Selon des cartes de variations de latitudes on trouve que la latitude de Laurentia était élevée pendant l'intervalle −2200 à −2000 Ma et de moyenne à basse pendant l'intervalle −1900 à −1500 Ma.

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Introduction

Because there are not many paleomagnetic studies of rock units older than 2 Ga (billions of years), many different polar paths have been proposed. In the earlier works (e.g. Spall 1971; Symons 1971, 1975; Irving *et al.* 1972; Fahrig and Chown 1973), the 2.6 to 1.8 Ga track (or part of it) is shown as an ascending one from present day Antarctica to north (30°) of the equator. Recently, a complete reversal in the sense of this path has been proposed (Irving and Lapointe 1975; Pullaiah and Irving 1975; Irving and McGlynn 1976*a*). Hence, Pullaiah and Irving have constructed for the period 2.2 to 1.8 Ga a path that runs from 70° N to 30° S. One major aberration in the construction of that path is the location of a well defined pole (mean 277° E, 18° S) obtained from the Nipissing diabase by several authors (Symons 1970, 1971; Patel 1971; Patel and Palmer 1974; Symons and Londry 1975). That pole appears to have received general acceptance as the pole at time of emplacement of the diabase, although this has not been proven or even claimed by some of those authors. Directions of magnetization different from those yielding that pole are also found in all the studies of the Nipissing diabase, showing that the magnetization is very complex; and it is far from certain, in the absence of thermal demagnetization work, which of the magnetizations (if any) was acquired at time of emplacement. Possibly, those magnetizations can be disentangled and a primary magnetization recovered by using both AF (alternating field) and thermal demagnetizations. Another alternative is to obtain the direction of the field from a contact test, since the Nipissing diabase intrudes the Huronian sediments. In this instance, it is highly desirable to determine the primary magnetization of the Huronian sediments in order to recognize if the magnetization at the contact is attributable to the emplacement of the diabase.

A pole (N) has been obtained recently from sediments of the Gowganda Formation in the Blind River – Elliot Lake area (Symons 1975), but the age of magnetization is uncertain. The results were obtained from alternating field (AF) demagnetization at 200 G (gauss; one specimen per site was demagnetized up to 800 G). No thermal or chemical cleaning treatment was performed to determine if one or more magnetizations were present. Without thermal and chemical work, it is difficult to establish the origin and probable age of magnetization(s) of sediments (Roy and Park 1974). The α_{95}s before (6°) and after (10°) correction for bedding tilt indicate that the magnetization was probably acquired after folding. The site-means estimate of precision κ (calculated from the α_{95}s) without and with correction for bedding tilt are 42 and 15, respectively. A F-distribution test for comparison of precisions (Watson 1956) gives κ uncorrected/κ corrected = 2.8, which is significant at the 99% level, indicating high probability that the magnetization was acquired after folding. This result (Pole N) will be discussed later with the other poles.

In 1973 and 1974, both the Huronian sediments and Nipissing diabase were sampled; the object of this work is four-fold: (1) to determine the direction of magnetization of the Huronian sediments at time of deposition, (2) to establish the direction of the field at time of emplacement of the Nipissing diabase by determining the direction of the PTRM (partial thermoremanent magnetization) of sediments in contact with the diabase, (3) to separate by AF and thermal work the primary and secondary magnetizations that appear to coexist in the Nipissing diabase, and (4) to construct from these new (directions of primary and secondary magnetizations of both Huronian sediments and Nipissing diabase) and previous data a polar path for the period −2.3 to −1.5 Ga.

Geology

Rocks of the Huronian Supergroup occur in the Southern Province as a belt 475 km long extending from Noranda, Quebec, to Sault-Ste-Marie, Ontario (Fig. 1). The Huronian Supergroup is divided into 4 groups: Elliot Lake, Hough Lake, Quirke Lake, and Cobalt. These rocks have been intruded by gabbro bodies of early Proterozoic ages, and in the Cobalt–Gowganda and Blind River – Sault-Ste-Marie areas, are referred to as the Nipissing diabase. Both the Huronian Supergroup and the Nipissing diabase have been regionally metamorphosed under conditions ranging from the lower greenschist to lower almandine amphibolite facies (Roscoe 1973). In the Cobalt–Gowganda area, the lower greenschist facies prevails.

Huronian Supergroup

In the Cobalt area, the Huronian Supergroup (Table 1) is represented by two formations of the Cobalt Group: the Gowganda and the over-

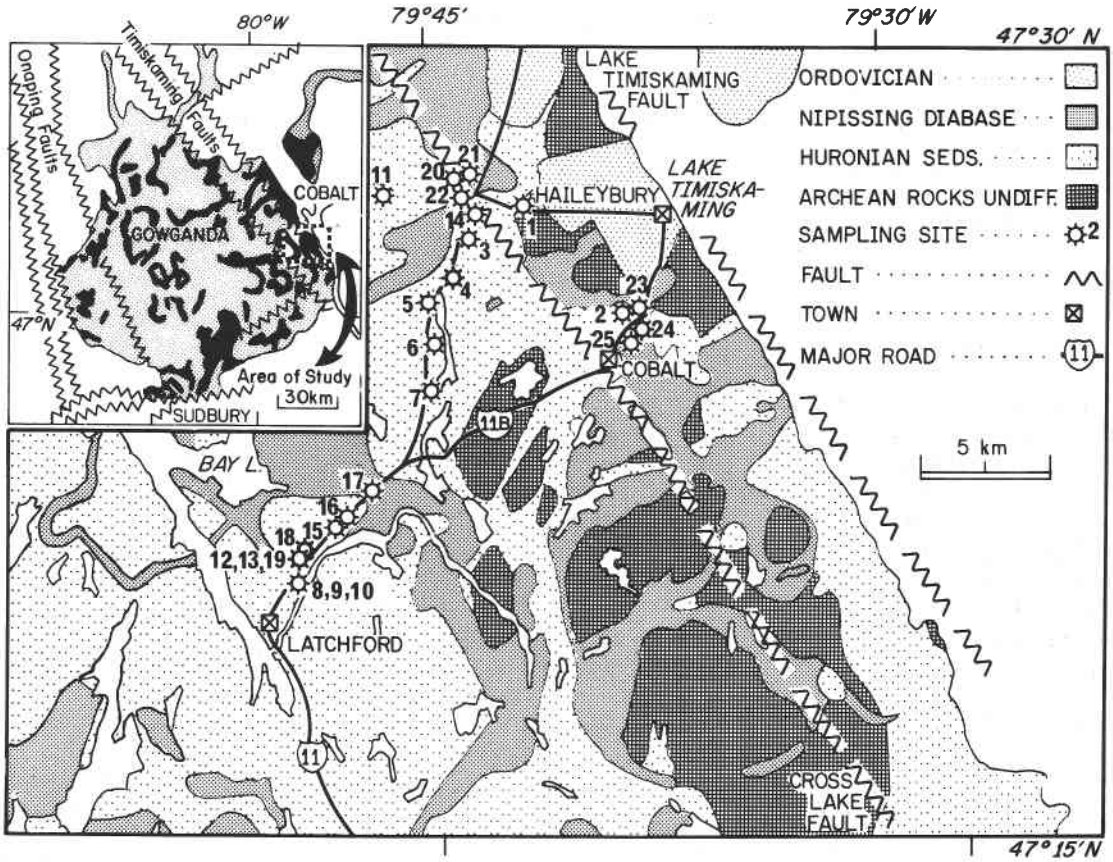


FIG. 1. Locality map showing the sampling sites.

TABLE 1. Stratigraphic sequence with sampling sites

ROCK TYPE	FORMATION	MEMBER†	SITES
Nipissing diabase			17-25
* * * * * * * intrusive contact * * * * * * * * * * * * *			
Cobalt Group (Huronian Seds.)	Lorrain		15-16
	Gowganda	Firstbrook	3-14
		Coleman	1-2

†The Coleman and Firstbrook nomenclature, as defined by Thompson (1957) and Jambor (1971), are the lower and upper Members of the Gowganda Formation. The Coleman Member consists largely of conglomerate with variable amount of argillite, quartzite, and arkose. The Firstbrook Member consists of well bedded, fine grained greywacke or argillite and this describes the sites sampled.

lying Lorrain. The Gowganda Formation (1000 to 1500 m thick) may be divided into two members (Thompson 1957; Jambor 1971), the Coleman Member and the overlying Firstbrook Member (Table 1). The Coleman Member consists primarily of conglomerate with variable amounts of argillite, quartzite, and arkose. The maximum thickness is about 180 m and the generalized upward succession starts from the basal conglomerate, through argillite, quartzite, and an upper conglomerate unit (Jambor 1971). The Firstbrook Member overlies the Coleman and consists of fine grained, well bedded argillite, arkose, and greywacke, some of which are reddish. The thickness varies from 210-305 m. An age of 2288 $\pm$ 87 Ma (millions of years) has been obtained from this formation by Fairbairn *et al.* (1969) (Table 2). The Lorrain Formation (1000 m thick) is composed of a lower unit of pinkish arkose, grading upward into a coarse orthoquartzite.

Nipissing Diabase

The rocks are intruded as vertical dikes or flat lying, sill-like intrusions and the contacts with the country rocks are invariably sharp. They consist of gabbro with differentiated phases of quartz gabbro, diorite, and granophyre. The

TABLE 2. Summary of isochron ages

Location	Rock name	Rb–Sr isochron* age (Ma)	Reference
Gowganda–Cobalt	Gowganda FM	2287 ± 88	Fairbairn 1969
Gowganda–Cobalt	Nipissing diabase	2162 ± 27	Fairbairn 1969
Elliot Lake	Nipissing diabase	2155 ± 80	Van Schmus 1965
From Sault Ste. Marie	Abitibi dike ENEW†	2147 ± 68	Gates 1971
To Lake Mistassini	Abitibi dike ENEN†	2262 ± 279	Gates 1971
	Abitibi dike NNE†	2227 ± 125	Gates 1971

*Decay constant used: $\lambda_{Rb} = 1.39 \times 10^{-11}\ y^{-1}$.
†Trend of dike (Larochelle 1966).

emplacement of the intrusions are thought to have been controlled by pre-existing faults and their distribution pattern by the different conglomeratic horizons; in the Cobalt Group, the Gowganda Formation is the stratigraphic controlling horizon (Card and Pattison 1973). From the Cobalt–Gowganda area, Van Schmus (1965) and Fairbairn *et al.* (1969) have obtained dates for the Nipissing diabase of 2155 ± 80 Ma and 2162 ± 27 Ma, respectively. The Cobalt–Gowganda area has been subject to fault movement after the Huronian deposition and Nipissing intrusions; the Murray and Onaping fault systems may originate from a major rift system and the Timiskaming from the St. Lawrence – Ottawa Valley rift system (Kumarapeli and Saull 1966).

Cleaning Methods

Three cleaning methods (thermal, chemical, and AF) are used; in addition a two-stage cleaning technique consisting of AF demagnetization followed by thermal demagnetization is also used. Thermal and AF demagnetizations were carried out using apparatuses operating in residual fields <2 gammas (Roy *et al.* 1969; 1972; 1973).

Thermal Demagnetization. All thermally treated specimens were demagnetized by heating to 100, 300, 400, 500, 550, 600, 650, and 675 °C. In many instances, 625 °C was also used.

Chemical Leaching. The treatment consisted of immersing the specimen (after drilling a 0.6 cm axial hole to increase the surface; Park 1970) in hydrochloric acid for the following accumulated times—25, 50, 115, 250, 370, 540, 1300, and 2300 h—washing in water, storing in zero field for two days, measuring, and re-immersing in fresh acid. The acid strength was 8 N for the first 370 h and 10 N thereafter.

AF Demagnetization. All specimens so treated were subjected to fields of 50, 100, 150, 200, 250, 300, 400, 500, 750, 1000, 1250, and 1500 G and subsequently several of them in steps of 250 to 3000 G.

Two-Stage Method. This method consists of thermally demagnetizing at 100, 400, 500, 550, 600, and 625 °C, specimens that had been previously AF cleaned in fields of 1500 to 3000 G. This method has been used successfully before (Park 1975; Irving and McGlynn 1975); based on the results obtained here, we shall present a hypothetical model to explain the fact that this two-stage cleaning technique permits one to recover discrete magnetizations that often cannot be uncovered by a one-stage cleaning technique. In Figs. 4 to 10, illustrating the results obtained by demagnetization, the treatments are identified as follows: thermal (triangles), chemical (squares), and AF (dots and circles). Closed (open) symbols indicate positive (negative) inclinations.

In the work on the Nipissing diabase, magnetizations with coercive forces <200 G are not specifically discussed, unless they seem related to one of the stable magnetizations. Soft magnetizations are present at most of the diabase sites; the directions within a site are usually near the present field direction, and sometimes randomly distributed; the intensities are large and often much (up to ten times) larger than the intensities of the stable magnetizations. It then becomes a serious problem to minimize spurious remagnetization of the grains carrying those soft magnetizations since, owing to the relative intensities, the slightest remagnetization may appear as an appreciable noise level superimposed upon the stable magnetizations remaining during an AF demagnetization in progressively higher fields, specially past the 1000 G field.

The measurements were performed using an astatic magnetometer (Roy 1971) and a Digital

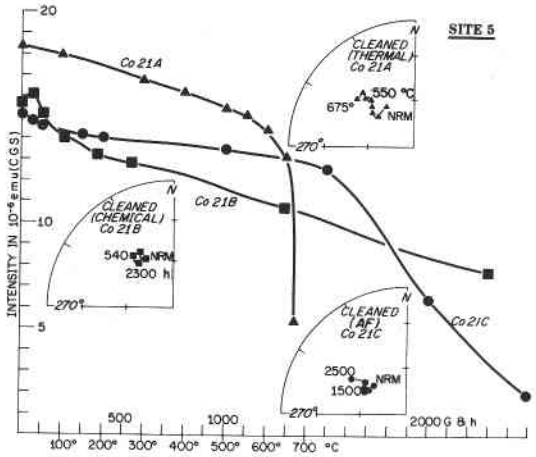


FIG. 2. Results obtained from thermal (°C), chemical (h), and alternating field (G) cleaning treatments, showing the stability of the initial magnetization (Grouping B) of the Firstbrook Member. The directions, all positive, have been corrected for bedding tilt. The magnetization is probably of chemical origin. The graduation below the abscissa indicates the temperature; that above indicates both the field in gauss and the leaching time in hours.

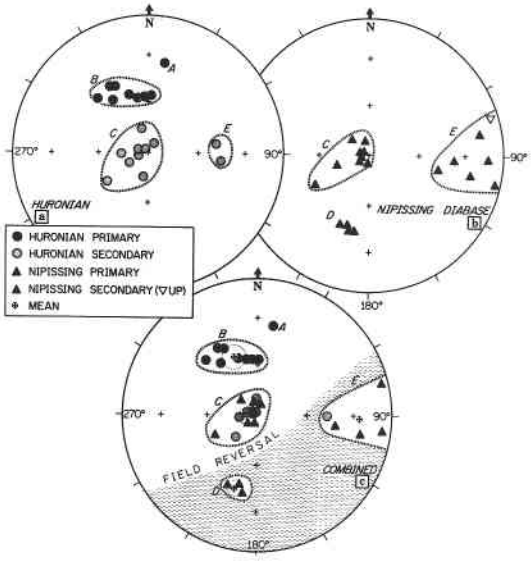


FIG. 3. Site means directions showing the 5 Groupings. In (a) and (b), all the site means obtained *either* from thermal *or* AF cleaning as listed in Table 6 are shown; (c) gives the site means obtained by averaging both thermal and AF results. The directions of Groupings D and E have been inverted so that all directions (except one) are shown with a positive inclination. The sequence of magnetization is from A to E and a field reversal evidently occurred between the times of magnetization of C and D. The direction of Groupings A and B are given after correction for bedding tilt, and those of Groupings C, D, and E are given with respect to horizontal.

Schonstedt spinner magnetometer (DSM-1), thereby minimizing possible systematic bias attributable to a measuring instrument (Park 1975).

Results

The sampling consisted of 178 cores (2.5 cm diameter) obtained at 25 sites (Fig. 1); 1, 2, or 3 specimens (height = 2.2 cm) were cut from each core depending on its length and a total of 257 specimens were subjected to cleaning. All from different cores, 121 specimens were thermally treated; the sites to which they belong are given in Tables 4 and 6. Also all from different cores, 111 specimens were subjected to AF demagnetization; 85 belong to the sites listed in Table 6, 3 to site 1 (Table 4), and 23 (11 of which are listed in Table 5) to sites 3–10. The two-stage method was performed on 48 specimens; 33 significant results were obtained, and those are listed in Table 6. Of 25 specimens subjected to chemical leaching, the results of 11 of them (a representative cross section of the chemical work) are in Table 5.

With the use of these different cleaning methods it is found that five different directions of magnetization are present. These directions have been assigned to five different groupings: A to E. Table 3 is a summary of the direction(s) found at each of the 25 sites sampled, together with the most probable age of magnetization (P or S) relative to the age of the rock unit. It is noted that both primary and secondary magnetizations are found in both the Huronian sediments and the Nipissing diabase. It is simpler to discuss the results mainly (but not exclusively) according to the grouping, since at many sites (20–25), two stable magnetizations coexist in the same specimens. Table 3 then becomes a useful reference.

Grouping A

Two sites were sampled from the Coleman Member, one (site 2) of which was a conglomerate in contact with the Nipissing diabase and which will be discussed later. Of the 7 specimens collected at site 1 (an argillite), 4 (3) were thermally (AF) treated (Table 4). From 20 to 350 °C, a change in direction (28° of arc) and a reduction in intensity (to 25%) occurred; similarly, from 0 to 300 G, a change in direction (24° of arc) and a reduction in intensity (to 40%) occurred. From 550 to 650 °C and from 300 to

1000 G, both the directions and the intensities remained essentially constant; directional (10°) and intensity (15%) changes occasionally occurred between individual readings, but most of those changes could be attributed to temporarily induced components. After cleaning in those temperature and AF ranges, the specimens at this site became very susceptible to any stray magnetic fields that they might encounter.

Because of the high blocking temperature and high RCF (remanent coercive force) spectra, it is believed that the magnetization remaining after cleaning was acquired shortly after deposition as discussed in the Grouping B results (below). It is stressed, however, that this is the result of only one site and it is unlikely that the secular variation has been averaged out. Therefore, the pole derived from Grouping A (assuming a dipole field) should be regarded not as an accurate, but as a rough estimate of the pole position during Coleman time.

Grouping B

Of the 12 sites collected from the Firstbrook Member (Table 3), three (sites 12, 13, and 14) were purposely sampled near exposed Nipissing diabase for a contact test. These and another site (11) with a similar magnetic behaviour are discussed later. All the specimens of the remaining eight sites (3–10) carry a single but slowly acquired magnetization as indicated by thermal, chemical, and AF cleanings (Fig. 2). More than 75% of the magnetization has blocking temperatures about 650 °C (Table 4); more than 60% of the magnetization remains after demagnetization in fields of 1500 G and the MDF (median destructive field) is about 2000 G. With such high blocking temperatures and high RCF, it is most probable that hematite is the magnetic

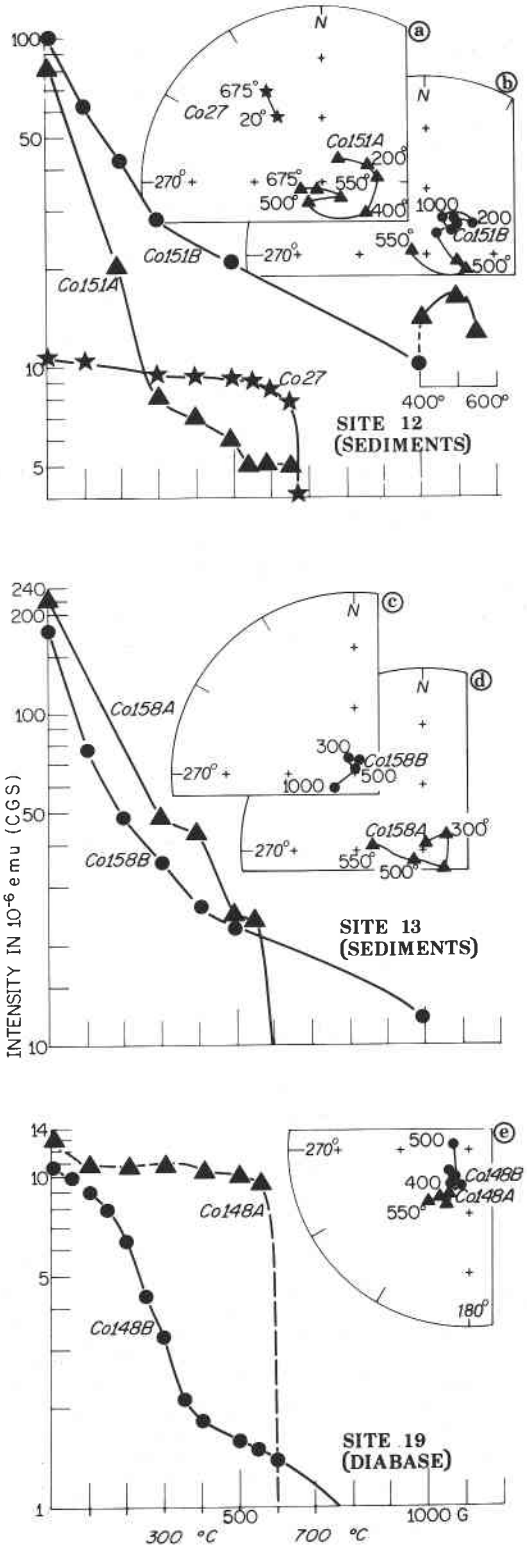


FIG. 4. Results from contact test giving Grouping C magnetization. The Firstbrook red beds of sites 12 and 13 abut the Nipissing diabase (site 19). The same direction of magnetization is found in both rock units, indicating that this magnetization was acquired at the time of cooling of the diabase. Both the direction and the decay curves of the magnetization at sites 12 and 13 are clearly different from those obtained in Grouping B (Co 27 of site 6 is shown for comparison). It is thought to be a TRM acquired after heating caused by the intruding Nipissing diabase. Co 151B shows results obtained by AF cleaning followed by thermal cleaning, a procedure often used in this study. In this and subsequent figures, curves ending on the abscissa but without symbols indicate the subsequent cleaning step used after the one yielding the last significant result.

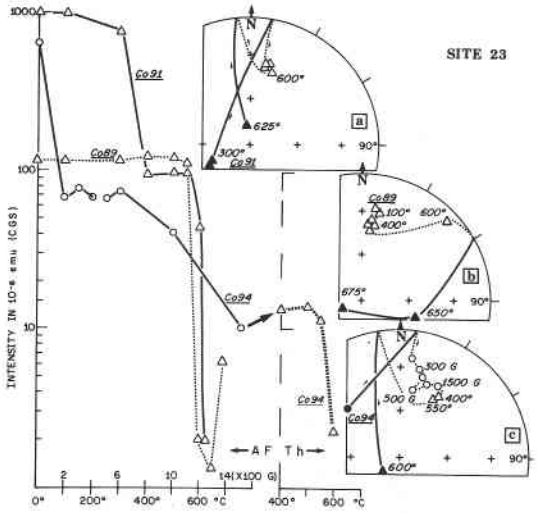


FIG. 5. Results showing that although the Grouping D magnetization is stable (up to 550 °C and 1500 G), it is superimposed upon an even more stable Grouping C magnetization (600–675 °C). This C magnetization cannot be recovered by AF cleaning alone (as shown in (c)).

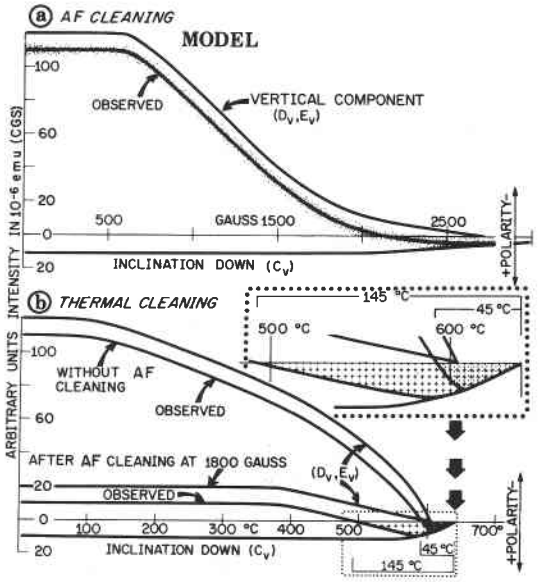


FIG. 6. A hypothetical model to explain the fact that the Grouping C magnetization can be best uncovered from under the D or E magnetization of greater intensity by using AF cleaning followed by thermal cleaning. For simplification, only the vertical components of C_v (positive polarity), and D_v and E_v (negative polarity) are used. With AF cleaning (a), the noise level of the initially very large intensity (see Cleaning Methods) is comparable to the small remaining intensity after AF demagnetization at 2200–3000 G. Although the D and E magnetizations have lower blocking temperatures than the C magnetization (b), the fact that their intensities are much larger limits the temperature range (45 °C) in which C dominates. Prior cleaning in high AF fields (1800 G) reduces the initial magnetization (to 17% in this model) and enlarges the temperature range in which the positive polarity magnetization dominates.

carrier. These argillites are very fine grained and well compacted, so that the penetration rate of the acid is very slow. Specimens were sliced at different stages of leaching for observation and there is a close correlation between the percentage of remaining intensity and the percentage of leached material. The penetration is clearly marked by a sharp change in colouring. After the 2300 h treatment, the leached part forms two buff concentric rings of 2–3 mm thickness enclosing a charcoal grey to reddish ring of 5 mm thickness, and the amount of unleached material is calculated to be about 40% of the total volume. The ratio of intensities after 2300 h/NRM is also about 40% (Table 5). It then appears that the chemical treatment has leached most (if not all) of the magnetic carriers from the leached part, while leaving the unleached part practically untouched. This is supported by examination of polished sections; no magnetic carrier could be detected in the leached part.

The direction changes occurring during cleaning are small, but different for each of the three cleaning treatments. Thermal demagnetization to 650 °C causes a systematic decrease in inclination (and a slight westwards change in declination) of 8 to 10° (Table 4, Fig. 2); AF demagnetization up to 1500–2000 G produces a decrease of about half that amount (Table 5, Fig. 2), whereas chemical leaching produces no appreciable direction change (Table 5, Fig. 2). There is no evidence that these specimens carry any soft viscous magnetization. Thermal demagnetization at 100 °C produces very little change in direction or intensity. These changes are not systematic among specimens and when a few results are averaged, there is no significant difference between the NRM and the 100 °C means (Table 5). Thus, it is concluded that no secondary magnetization is present and that all of the magnetization (NRM) was acquired during the initial magnetization process. This is in accordance with a study of polished sections, described later.

These results are comparable to the results obtained from Carboniferous red beds and a similar process of magnetization is proposed

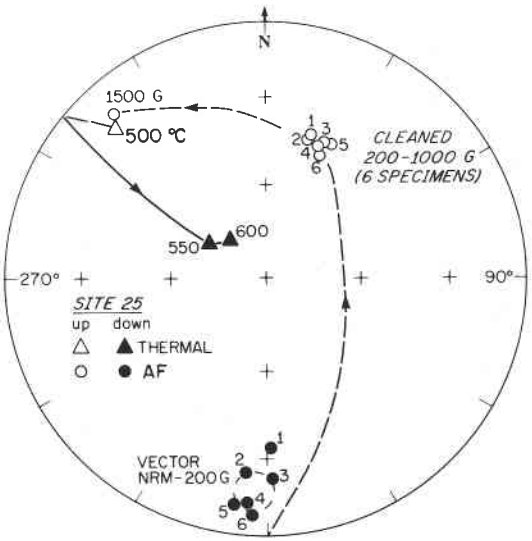


FIG. 7. Directions of 6 specimens obtained by AF treatment. The D magnetization is clearly shown in the northeast and up direction. In all specimens, this direction was reached at 200 G and remained there up to 1000 G. The southwards direction was obtained by vector analysis and the tight grouping suggests that it may be a reversed D direction. The underlying C magnetization was recovered by subsequent higher AF and thermal cleaning in all 6 specimens (Table 6) and one example is shown.

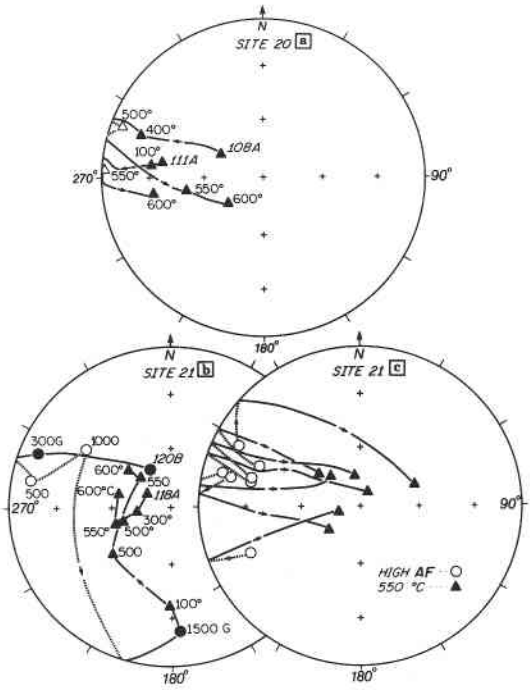


FIG. 8. Results showing Grouping E magnetization superimposed upon the C magnetization. Under thermal cleaning, the presence of the E direction is revealed at about 500 °C in most (a), but not all (b—118A) instances; and by AF cleaning (b, c) in fields up to 1000 G (Table 6). The C magnetization (steeply down) could be isolated by thermal cleaning after AF cleaning (c) and, except for a few exceptions (111A in a), by thermal cleaning at 600 °C (a, b).

(Roy and Park 1974). The process began with the formation of hematite nuclei at or shortly after deposition, followed by hematite growth, filling, and cementing during diagenesis. In that study it was shown, by combining thermal and chemical work, that the magnetization carried by the red pigmentation and therefore acquired during the late phases of magnetization had blocking temperatures distributed over a wide range from 20 to 550 °C (and up to 674 °C for some specimens), while the magnetization acquired during the early phases had blocking temperatures confined to a short interval below the Curie point of hematite. Because of the difference in porosity, a significant comparison of chemical results between the well compacted argillites and the more porous Carboniferous red beds cannot be made. However, the thermal results obtained in the two studies are very similar and it is reasonable to suggest that the magnetization (Fig. 2) remaining after demagnetization at 550–650 °C was acquired during the early phases of magnetization and that the magnetization with blocking temperatures below 550 °C is carried by the red cement and was acquired during the late phases. Therefore, we

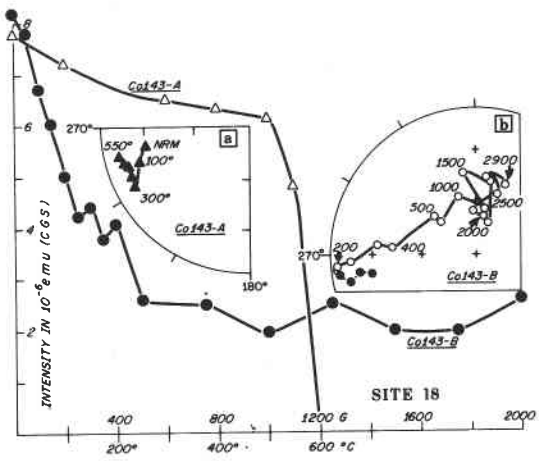


FIG. 9. Specimen carrying both the D and E magnetizations. The fact that D has the higher remanent coercive force spectrum suggests that it predates E.

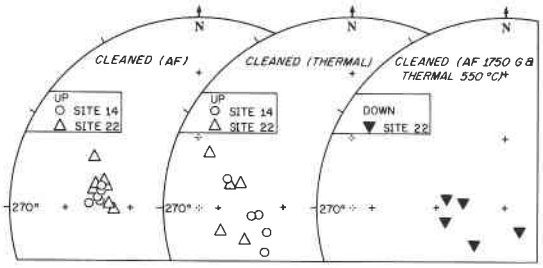


FIG. 10. Results from an argillite (14) and a diabase (22), sites separated by a fault. Because of the agreement between directions (a, b), the E magnetization is associated with faulting. The C magnetization could be recovered by AF followed by thermal cleaning at site 22, but not at the argillite site 14, which maintained the E direction up to 600–650 °C.

distinguish the direction obtained after thermal demagnetization (Table 4; 550–650 °C) as representing the direction of the magnetization acquired at or shortly after deposition and the direction given by the NRM (or 100 °C; Table 4) as representing the mean direction of the magnetization acquired during the whole magnetization process. Forty per cent of the specimens were demagnetized at 675 °C. The intensities were 20–30% of those of the NRM and the directions were usually within 4° to 5° of the 650 °C measurements. Accurate measurements were not easily obtained owing to temporary induced magnetizations[2], probably acquired by the demagnetized carriers. However, the results indicate that for most specimens, a good portion of the magnetization has blocking temperatures at least as high as 675 °C.

Although all phases of magnetization have high RCF spectra (there are no appreciable changes in direction or intensity up to 1000 G: Fig. 2), they can be partly separated in high alternating fields. Indeed, above 1500 G, the intensity decreases and the direction changes occurring then are in the same sense as those taking place during thermal demagnetization (Table 5); this indicates that the magnetization of the early phases has higher RCF than that of the late phases. Yet, the phases could not be separated chemically. This is attributed to the slow rate of penetration of HCl, which allows enough time for the acid to destroy all magnetic carriers (regardless of the grain size and solubility) before proceeding to attack any of the grains buried deeper within the specimen. This means that any remaining portion of unleached material still represents the mean direction of

TABLE 3. Sites sampled from the Huronian sediments and the Nipissing diabase

Site	Magnetization	Grouping
Coleman Member		
1	P	A
2	S	C
Firstbrook Member		
3	P	B
4	P	B
5	P	B
6	P	B
7	P	B
8	P	B
9	P	B
10	P	B
11	S	C
12	S	C
13	S	C
14	S	E
Lorrain Formation		
15	S	C
16	S	?
Nipissing Diabase		
17	P	C
18	S	E
19	P	C
20	P, S	C, E
21	P, S	C, E
22	P, S	C, E
23	P, S	C, D
24	P, S	C, D
25	P, S	C, D

P and S are primary and secondary magnetizations, respectively, as defined in the text.
Five distinct magnetizations can be found in these rocks. Each direction has been assigned to a Grouping (A–E).

[2]These induced magnetizations could readily be observed with the astatic magnetometer because of the use of the direct read method. The direction of magnetization is obtained when it is parallel to the direction of the magnet system; thus, the error introduced by a short term induced magnetization is negligible, but at the same time the determination of the remanence direction is difficult and laborious because the magnetometer responds to the induced magnetization as if it is in an unstable mode. With the spinner magnetometer, this induced magnetization is translated into a large increase in integrating time (slower measuring speed). The intensities measured tend to be larger (0 to 40%) than those measured with the astatic; these larger intensities may indicate that part of the induced magnetization is integrated. Directions measured on the two magnetometers are in good agreement, except occasionally when the difference in intensity is large. Comparison of the most reliable 675 °C results with the 650 °C results shows no systematic difference in direction except for perhaps an inclination 2° lower than that given in Table 4.

TABLE 4. Analysis of intensities and directions after thermal and AF cleaning of Huronian Sediments (Gowganda Formation)

Site	N	Mi	$D°,I°$	R	Mi	$D°,I°$	R	κ	Poles Lat. °N	Poles Long. °E
Group A (Coleman Member)										
		NRM			550–650 °C					
1a	4	10.0	349, +52	3.8870	2.4	019, +33	3.9356	47	57	065
					300–1000 G					
1b	3	9.8	345, +58	2.9088	3.9	358, +34	2.8328	12	62	095
1	7					010, +34	6.6872	19	60	080
Group B (Firstbrook Member)										
		100 °C			550–650 °C					
3	4	11.2	347, +60	3.9396	8.6	341, +53	3.9373	48	70	153
4	13	15.1	328, +58	12.9017	12.5	324, +52	12.9069	129	59	175
5	6	17.1	319, +54	5.9682	14.4	315, +45	5.9662	148	49	176
6	6	9.1	340, +55	5.7965	7.4	329, +43	5.9598	124	56	159
7	6	9.9	334, +53	5.9521	7.9	332, +45	5.9656	145	60	157
8	9	9.1	005, +60	8.9707	7.4	356, +56	8.9673	245	79	117
9	7	9.1	011, +62	6.9616	7.4	002, +56	6.9641	167	79	092
10	5	10.0	358, +63	4.9831	8.4	349, +56	4.9825	229	76	141
MEAN	8	11.3	344, +59	7.8890	9.3	337, +52	7.8585	49	67	158
				$\alpha_{95} = 7.0°$		$\alpha_{95} = 7.9°$				(7.8°, 11.2°)*

N = number of specimens; Mi = intensity in 10^{-6} G; R = resultant vector; κ = precision estimate (Fisher 1953); α_{95} = radius of circle of confidence ($P = 0.05$).
550–650 °C means that each specimen has been demagnetized at 550, 600, and 650 °C; the average of these three results has been used as unit weight.
300–1000 G is the average of AF cleaning at 300, 500, and 1000 G.
*Semi-axes (δp, δm) of the oval of confidence ($P = 0.05$).

magnetization acquired during the entire magnetization process and explains the fact that no appreciable changes of direction occur during leaching (Table 5). The small changes, not unidirectional amongst specimens (Table 5, Fig. 2), are probably attributable to secular variations during the magnetization process. Both the magnetic results and the observations of polished sections (described later) indicate that all the magnetic carriers can be leached out; it then appears that all of the magnetization is of chemical origin.

None of the cleaning treatments gives any indication that a field reversal occurred during the magnetization process. This is important since it has been shown that direction of magnetizations acquired at field reversal times can easily be inaccurate and unreliable (Roy and Robertson 1968; Roy and Park 1969; Larson and Walker 1975), unless the different phases of magnetization can be separated by some experimental and analytical method (Roy and Park 1974).

All beds are within 7° of the horizontal and the results given are after correction for bedding. (Without correction, the direction is 340°, +56°.) The standard deviation of the site is 11.5° and comparable to the 13° calculated as the direction dispersion of the present field at that latitude (Creer 1962). Assuming a secular variation comparable to the present one, this indicates that the secular variation has been averaged out and that the overall mean (337°, +52°; Table 4) represents the direction of the field at that time. Because of the small difference ($\simeq 10°$) between the 100 °C and the 650 °C results, it is suggested that the process of magnetization was short and/or that the apparent displacement of the pole during the magnetization process was small.

Grouping C

The sites giving directions represented by this grouping were obtained from both the Huronian sediments and the Nipissing diabase (Fig. 3 and Table 6). Sites 2, 12, and 13 were sampled because they were in contact with the Nipissing diabase. Sites 12 and 13 from the Firstbrook Member were apparently pushed up and tilted

TABLE 5. Comparison of results of paired specimens subjected to chemical and thermal treatments (first part), and to AF and thermal treatments (second part)

		NRM		2300 h				100 °C		650 °C	
Site	Specimen	*Mi*	*D°,I°*	*Mi*	*D°,I°*	Ratio*	Specimen	*Mi*	*D°,I°*	*Mi*	*D°,I°*
3	2A	14.3	349, +57	7.6	351, +59	0.53	2B	10.2	018, +58	7.7	356, +53
4	9C	16.5	317, +61	3.8	320, +63	0.23	9A	12.7	327, +62	9.3	320, +50
5	20C	16.8	321, +59	5.8	327, +57	0.35	20A	15.3	321, +60	12.3	311, +46
5	21B	14.8	326, +50	7.8	320, +48	0.53	21A	18.0	317, +57	14.5	317, +44
6	25B	6.6	305, +56	2.8	317, +60	0.43	25A	11.4	324, +44	9.6	318, +37
6	26A	10.8	333, +48	5.2	329, +51	0.48	26B	10.0	330, +48	8.6	332, +40
7	29B	11.8	317, +44	4.3	322, +45	0.37	29C	10.7	332, +53	8.4	331, +40
7	32B	11.2	329, +53	5.1	329, +54	0.46	32A	10.9	334, +45	9.7	334, +41
7	35A	10.2	331, +53	4.7	338, +56	0.46	35C	9.7	337, +50	7.6	330, +40
8	46A	8.2	334, +56	5.2	351, +57	0.57	46C	9.8	348, +57	7.9	345, +46
9	61	7.8	000, +52	3.4	359, +51	0.44	60	8.6	006, +54	6.8	337, +50
MEAN	11	11.7	331, +55	5.0	333, +55	0.43		11.6	337, +55	9.3	335, +45
(κ)			(53)		(66)				(43)		(41)
					1500 G						
3	7B	12.3	354, +57	8.1	352, +55		7A	9.7	338, +55	7.0	339, +41
5	21C	15.2	311, +64	12.6	308, +58		21A	18.0	317, +57	14.5	317, +44
6	24B	8.3	330, +59	4.5	335, +49		24A	7.0	327, +60	5.5	338, +55
7	35B	9.4	326, +58	6.7	328, +50		35C	9.7	337, +50	7.6	330, +40
8	46B	9.9	347, +59	5.0	349, +55		46C	9.8	348, +57	7.9	345, +46
10	176	11.7	009, +57	8.1	357, +62		175	8.2	001, +59	6.2	349, +53
10	178	10.8	008, +61	7.0	001, +62		177	10.6	347, +58	6.5	349, +46
MEAN	7	11.1	344, +61	7.1	341, +57			10.4	340, +58	7.9	338, +48
(κ)			(53)		(51)				(79)		(62)

*This is the ratio of the intensity remaining after 2300 h of leaching over the NRM intensity.

TABLE 6. Site-mean directions after thermal and/or AF treatments of all sites (except those listed in Table 4). The Grouping (C, D, or E) to which the direction belongs is indicated

Site	Temp. °C†	Thermal treatment N	Grouping	D°,I°	R	κ	Field (G)†	AF treatment N	Group	D°,I°	R	κ
Huronian sediments												
2	625	5	C	035, +84	4.6165	10	300–1000	3	C	345, +75	2.9937	318
11	300	5	C	240, +78	4.8799	33						
11	550	4*	C	234, +60	3.1371	4	500	4	C	191, +75	3.9475	57
12**	500–625	7	C	305, +82	6.6325	16						
12	*500–625*	*7+3**	*C*	*286, +84*	*9.5556*	*20*						
13	*500–600*	*8*	*C*	*269, +74*	*7.0315*	*7*	200–1000	6	C	256, +84	5.8620	36
14	600–650	5	E	262, −48	4.9327	59	300–500	5	E	277, −45	4.9839	248
15							100	10	C	334, +88	9.8105	48
16							100	7		300, −55	6.8355	37
Nipissing diabase												
17	600–650	6	C	192, +85	5.6367	14	1250	3		008, −25	2.553	4
18	500	3	E	253, +06	2.8888	18	300–1250	4	E	260, −18	3.8316	18
19	550	3	C	240, +53	2.9905	211	*200–500*	*4*	*C*	*253, +70*	*3.9072*	*32*
20	500	2	E	283, −04	1.9745	—	300–1000	6	E	278, −12	5.8967	48
20	550–600	2+6*	C	283, +85	7.2903	10						
21**	550	2	C	251, +63	1.9985	—						
21	550	2+7*	C	311, +77	8.4441	14	200–1000	8	E	281, −24	7.7417	27
22	550	5	E	274, −36	4.8428	25	300–1500	7	E	281, −46	6.9443	108
22	550	5*	C	277, +85	4.7113	14						
23	500–550	5	D	014, −42	4.9671	122	300–1000	5	D	011, −42	4.9852	271
23	600–675	3+1*	C	345, +82	3.8015	15						
24	600–675	3+1*	C	289, +87	3.7429	12	300–1000	7	D	017, −45	6.9177	73
25	600	2+6*	C	235, +83	7.3278	10	200–1000	6	D	022, −44	5.9912	606
MEAN		13	C	260, +82	12.6717	37		5	C	260, +83	4.8986	39
MEAN		1	D	014, −42	—	—		3	D	017, −44	2.9944	357
MEAN		4	E	268, −21	3.6334	8		5	E	274, −29	4.8153	22

*Thermal treatment after AF demagnetization in fields (f); 1500 < (f) < 3000 G.
**Not used in means because included in next entry.
†The meaning of the two cleaning values is that within that range the direction changes were negligible. Therefore the direction listed is the mean of directions obtained at each of the cleaning steps given in 'Cleaning Methods'.
Boldfaced characters indicate the means and contact results estimated as the most reliable.
Sites 2–16 are from the Huronian sediments and 17–25 from the Nipissing diabase.
Sites 16 and 17 AF have not been assigned to a grouping because of reasons given in the text.
All directions given in this table and corresponding figures are with respect to the horizontal.

by 12° at 130° DDA (down dip azimuth) and by 35° at 145 DDA respectively during intrusion of the diabase, which was also sampled at that location (site 19).

Analysis and comparison of the intensity decay curves at these sites (Fig. 4) with those of Grouping B sites (Fig. 2 and specimen Co27 of Grouping B in Fig. 4) clearly indicate magnetizations of different origins. The NRM intensities are much higher (factors of 10 to 24) than those of the Grouping B specimens, and the decay curves with blocking temperatures distributed from 20 to 650 °C are in sharp contrast with the square shouldered curves of Grouping B. The AF decay curves also show a marked difference between these sites with most of the RCF below 1000 G and the Grouping B sites with RCF above 1500 G. Furthermore, there is no indication of the direction found in the Firstbrook Member (Table 4), suggesting that the sediments were heated above their Curie point, thereby removing the initial magnetization. The shape of the decay curves suggests a slow rate of cooling and the direction changes (Fig. 4a,d) are not unlike secular variation changes (Thellier 1966). It is suggested that this is a TRM (thermoremanent magnetization) acquired at the time of emplacement and cooling of the Nipissing diabase. The square-shouldered demagnetization curve (Fig. 4e) of the adjacent site 19 (Nipissing diabase) indicates that a single magnetization is present; the agreement between its direction and the direction obtained at sites 12 and 13 suggests that it was acquired during cooling, and that it is the primary magnetization of the Nipissing diabase. The mean of these 3 sites gives a direction of 264°, +76° with a precision estimate $\kappa = 103$. A second contact test between a Coleman conglomerate (site 2) and the adjacent diabase site 23 also give similar results (Table 6). This steeply inclined magnetization can be recovered at all argillite sites listed in Table 6, except site 14; since the magnetization at these sites is secondary, the directions are given with respect to the horizontal, that is, without correction for bedding; only sites 12 and 13 were appreciably off the horizontal.

At site 11, the thermal and AF (and AF + thermal) experiments clearly show that the C magnetization is present. After the 550 up to the 650 °C treatment, the direction moved towards the Grouping B direction, suggesting that the primary magnetization may not have been removed completely by heating at the time of the diabase intrusion. However, the specimens became extremely susceptible to induced magnetizations (see previous footnote) and the presence of the B magnetization could not be established with certainty. The C magnetization at this site is probably a PTRM attributable to the emplacement of the Nipissing diabase, although the nearest exposed diabase is one kilometre away. An unexposed diabase could, of course, be much nearer.

The Grouping C magnetization could also be recovered from all the Nipissing diabase sites but site 18. Although other stable magnetizations (Grouping D and E) are present at most of those sites, the C magnetization is found to be the most stable. It can often be uncovered by thermal magnetization in the 550–650 °C range (Fig. 5a,b). It can also be sometimes detected but rarely isolated by AF demagnetization in the 2000–3000 G range. In that range, the intensities were much less than 1% of the original NRM and it appears that noise originating from spurious magnetizations (probably acquired by the $99^{+}\%$ of the demagnetized carriers; see cleaning methods) and possibly small anhysteritic components prevented accurate determination of the direction of magnetization. However, it was found experimentally that this magnetization could be recovered much more often if the specimens were first demagnetized in 1500–2000 G fields and then subjected to thermal demagnetization (Fig. 5c).

In order to explain this, the following points can be made: (a) Any of three magnetizations (Groupings C, D, and E) can be found in the Nipissing diabase. C has a positive polarity; D and E have a negative polarity. (b) It is probable that at least two magnetizations are carried by hematite–ilmenite intergrowth; this is indicated by the fact that the RCF are usually >1000 G and the blocking temperatures range from 550 to 650 °C. (c) The D and E magnetizations have blocking temperatures slightly lower than the C magnetization (since the latter can often be separated by thermal demagnetization). (d) The RCF of C are larger than those of D or E; C can sometimes be detected by AF cleaning in 2000–3000 G. (e) The D and E remanent intensities are usually much larger than the C intensities. (f) Magnetization C is more often recovered when using a two-stage (AF followed by thermal) than a single stage demagnetization technique.

TABLE 7. Grand means and poles for each grouping and CONTACT, calculated from Table 6 data

		N	Grouping	D°, I°	R	κ	α_{95}	Pole position Lat. °	Long. °E	(δp, δm)
Mean	**(a)**	**18**	**C**	**259, +82**	**17.5929**	**42**	**5.4**	**42N**	**258**	**(11, 11)**
MEAN	(b)	13	C	269, +83	12.8261	69	5.0	45N	261	(10, 10)
MEAN	(a)	4	D	016, −43	3.9936	472	4.2	16S	265	(05, 06)
Mean	**(b)**	**3**	**D**	**017, −44**	**2.9960**	**503**	**5.5**	**15S**	**264**	**(05, 07)**
Mean	**(a)**	**9**	**E**	**271, −26**	**8.4196**	**14**	**14.4**	**09N**	**000**	**(09, 15)**
MEAN	(b)	5	E	272, −25	4.7373	15	20.2	08N	000	(12, 22)
CONTACT	**CO_1**	**3**		**264, +76**	**2.9806**	**103**	**12.2**	**39N**	**246**	**(24, 24)**
CONTACT	CO_2	3		257, +76	2.9575	47	18.2	36N	248	(35, 35)
CONTACT	CO_3	8		277, +81	7.7315	26	11.0	47N	255	(21, 21)

(a) Mean obtained by using all (thermal and AF) site means for that grouping.
(b) Mean obtained by using the mean for each site (of that grouping) after averaging the thermal and AF site means.
CO_1 is the mean of 12 (Thermal), 13 (Thermal), and 19 (AF) (italicized).
CO_2 is the mean of site means 12, 13, and 19 using both thermal and AF results and specimen unit weight.
CO_3 is the mean obtained by using both thermal and AF site means of 12, 13, and 19, and also those of the second contact: sites 2 and 23.

A model that is consistent with these results and observations is illustrated in Fig. 6. For simplification, only the vertical component of C, D, and E are considered. Under AF cleaning (a), the positive polarity may be detected between 2200 and 3000 G, but because of the noise level, it is difficult to determine the direction of magnetization. Under thermal cleaning (b), the positive polarity is only visible through a 45 °C window and could easily be missed when using a stepwise demagnetization technique. The effect of prior cleaning at 1800 G is to increase the width of the window to 145 °C (with a slightly larger intensity) and thereby to optimize the availability of the positive polarity magnetization. Two remarks can be made about this model: (*a*) the decay curve may be more (less) square-shouldered, producing a smaller (larger) window; and (*b*) the choice of the alternating field may be critical: too high a field (say 3000 G) may destroy the C_V magnetization; too small a field (say 800 G) may have no effect.

The two-stage technique was applied to 48 specimens. The final AF treatment of those specimens varied from 1500 to 3000 G. The C magnetization was recovered (at the indicated temperature—Table 6) from 33 specimens, most of which had been cleaned in 1500–2000 G; for the remaining 15 specimens, most of which had been cleaned in 2000–3000 G, the results were non-repeatable and meaningless, owing to a low signal-to-noise ratio.

It is noted that at some sites, κ is not large. This is to be expected, since 2 or 3 very differently oriented stable magnetizations coexist in the same specimens and it is improbable that those magnetizations have been completely separated in all specimens. Incomplete removal of oppositely directed magnetizations can produce large increases of within-site dispersion (Roy and Park 1969). Nevertheless, agreement amongst site mean directions obtained from Nipissing diabase, Huronian sediments, and contacts (Fig. 4) leaves little doubt that the Grouping C magnetization was acquired during the cooling of the Nipissing diabase and that the mean of Grouping C (Table 7; $\kappa \simeq 40$) provides a good estimate of the field direction at that time.

Grouping D

The direction of magnetization defined by this grouping has been clearly established by several workers (see Addendum A) from AF demagnetization experiments. From studies of polished sections and because of high (1200 G) coercive forces, Patel and Palmer (1974) believe that this magnetization is probably carried by ilmenite–hematite intergrowth. This is in accord with our findings. This magnetization is present in 3 sites: 23, 24, and 25. In general, neither the direction ($\simeq$017°, −44°) nor the intensity (1×10^{-4} emu) changed appreciably in progressively higher demagnetizing fields from 300–1000 G (Fig. 5c; Table 6). The blocking temperatures were generally in the range above 600 °C (Fig. 5a). Unquestionably, this magnetization is stable and its direction can be accurately determined. It is then a valuable marker of the field direction at the time of acquisition. However, it is doubtful that it was acquired at the time of emplacement because of the coexistence of the thermally more stable C magnetization (discussed previously) with blocking temperatures of 600–675 °C.

The AF treated specimens of sites 23 and 24

were demagnetized in fields up to 3000 G. In most instances, above 1500 G the signal/noise ratio is small and no meaningful results could be obtained. However, in some specimens, there was a tendency for the directions to move southwards in the 2000–2500 G range and one specimen gives a definite direction of 165°, + 42° and 169°, +34° at 1750 and 2000 G, respectively (near the reverse polarity of the D direction). It is therefore possible that a reversal occurred at acquisition time of the D magnetization. This possibility is enhanced by the findings at site 25 (Fig. 7); a vector subtraction shows a good grouping ($\kappa = 60$) not far away (26°) from the reverse D direction. Evidence for a reverse D direction has also been found in other studies (*e.g.*, Symons and Londry 1975).

The D magnetization is most probably present at site 17 also, and the behaviour of the direction indicates that this magnetization may have been acquired during a reversal. The directions first move towards the north at 1250 G and then in higher fields they move southwards with low inclinations. The direction of the normal or reverse polarity could not be determined with accuracy and therefore the direction at this site (Table 6) was not included in the computation of the D mean direction. Its inclusion would have produced a mean of 014°, −39° (pole: 267° E, 19° S).

Grouping E

This direction is often found at sites sampled near faults that occurred after emplacement of the Nipissing diabase. The magnetization has coercive forces usually requiring high destructive AF fields (1000–2000 G) and blocking temperatures of 550 °C for diabase and 600–650 °C for argillite (Table 6). At three sites (20, 21, and 22) this magnetization is superimposed upon the C magnetization; the latter can be uncovered by thermal treatment generally with (Fig. 8b,c) and sometimes without prior AF cleaning (Fig. 8a,b).

The E magnetization is probably younger than the D magnetization. At most of the sites (where the E magnetization is present), some specimens showed a directional change tending towards the D or reverse D direction (Fig. 8b); and at site 18 (Fig. 9), a specimen definitely shows a directional change from the E direction to the D direction.

The argillite site 14 and the Nipissing diabase site 22 were sampled across and adjacent to a fault and can be considered as a fault contact test (Fig. 10). The agreement between the direction obtained from both AF and thermal experiments shows that it is reasonable to attribute this magnetization to consequences (*e.g.*, heating, chemical alteration) of faulting.

Lorrain Formation Sites

Two sites (15 and 16) were collected from the Lorrain quartzite. The weak NRM (intensities $<0.8 \times 10^{-6}$ emu) were scattered ($\kappa = 4$ and 2, respectively). They were AF cleaned in fields of 25, 50, 75, 100, and 125 G. The best grouping ($\kappa = 48$ and 37) occurred at 100 G, with intensities of 0.1 to 0.3×10^{-6} emu. The results have been listed in Table 6, although they may be of no great significance because of the low cleaning field. Site 15 appears to belong to Grouping C; its inclusion or exclusion is unimportant for the Grouping C mean direction. Site 16 has not been assigned to any grouping. Its mean direction 300°, −35° (pole = 326° E, 09° N) lies between the D and E directions.

Polished Section Petrology

Huronian Sediments

In the unleached argillites, two possible magnetic carriers were detected: one is a very fine hematite pigment forming a thin film around minerals and producing the red coloration of the rock, and the other consists of small anhedral grains of hematite scarcely ($\simeq 0.05\%$) and homogeneously distributed throughout the rock. The specimens examined are fracture-free and there is no evidence of any alteration (capable of causing a secondary magnetization) that could have taken place after lithification. In the leached section of the chemically treated specimens, no magnetic carrier could be detected.

Nipissing Diabase

At most sites, two magnetic carriers were recognized: one is hematite in composite hematite–ilmenite and the other is magnetite in composite magnetite–ilmenite. Samples from sites 23, 24, and 24, where both the C and D magnetizations are clearly present (Tables 3 and 6), were examined in more detail. The hematite–ilmenite can be described as an ilmenite grain containing hematite lamellae of variable length and width 2–4 μm. Another possible carrier consists of anhedral grains of titanomagnetite with thin ilmenite lamellae (5 μm in length, 1–2 μm in width). The titanomagnetite is relatively fresh-looking and always occur in the vicinity of the hematite–ilmenite. These two carriers are also

found at other sites, but not necessarily near to each other.

Because the distribution, ratio, and ordering of hematite–ilmenite and ilmenite–magnetite vary greatly amongst samples, a simple correlation between the magnetic carriers and the different magnetizations cannot be established. In fact, it is quite possible that the different magnetizations might be carried by different phases of hematite–ilmenite and/or ilmenite–magnetite at different sites. Furthermore, in most instances, it is not clear whether the magnetizations are thermoremanent or chemical remanent. The presence of both hematite and magnetite, the high blocking temperatures, and high RCF indicate that the primary and the two secondary stable magnetizations could be TRM or CRM. A secondary TRM or PTRM could result from a reheating event or burial; a secondary CRM could be produced during a period of metasomatism. A primary TRM could be acquired during initial cooling of an igneous body, and there are numerous examples given in the literature. A primary CRM is much less discussed, although studies by a few workers (*e.g.* Buddington and Lindsley 1964; Ade-Hall *et al.* 1965; Carmichael and Nicholls 1967; Grommé *et al.* 1969) indicate that this type of magnetization might be of frequent occurrence. For example, Buddington and Lindsley have shown that, in many rocks, some phases in the FeO–Fe_2O_3–TiO_2 system do not reach equilibrium until cooling to temperatures as low as 650–550 °C. Since this is the range of blocking temperatures for hemoilmenite and titanomagnetite, the primary magnetization could easily be a CRM. Deuteric alteration at low temperatures could also produce CRM. Evidence for primary CRM might be found at some of the sites where the blocking temperatures of the primary magnetization C are clearly above the Curie point of magnetite, and range from 600 up to 675 °C in some instances.

Synopsis of Results

The site mean direction obtained from the 25 sites sampled form five different groupings (Fig. 3). Primary and secondary magnetizations are found in both Huronian sediments and Nipissing diabase (Table 3), and at some sites they coexist (Table 6).

Grouping A. The magnetization was acquired at or shortly after deposition based on thermal and AF experiments. The direction obtained from a single site can only be considered as a very *rough* estimate of the field direction during Coleman time.

Grouping B. Chemical, thermal, and AF treatments indicate that the magnetization is mostly of chemical origin and was acquired during the initial magnetization process that commenced with deposition of the beds. It appears that about 20% of the magnetization was acquired in the late phases, and that the magnetization (80%) remaining after thermal demagnetization at 650 °C was acquired during the early (but not necessarily the first) phases. By analogy with previous results (Roy and Park 1974), it may be suggested that the magnetization occurred within a few tens of Ma of deposition, meaning that if the dating of Fairbairn *et al.* (1969) is accepted, the magnetization is probably older than 2200 to 2250 Ma. This is a reasonable assumption according to the relative positions of poles of that age as discussed later. Standard deviation indicates that the sampling was adequate and that the mean direction is a good estimate of the field direction of that time.

Grouping C. This direction is obtained from both Huronian sediments and Nipissing diabase sites. The magnetization characteristics determined by thermal and AF treatments of the Huronian sediments sites are clearly different from those of the Grouping B sites, indicating magnetizations of different origins. Two contact tests indicate that the secondary magnetization is probably a TRM attributable to the emplacement of the Nipissing diabase. This is supported by the fact that a discrete but extremely stable magnetization (blocking temperatures of 550–625 °C) is found in all but one of the Nipissing diabase sites. The mean direction obtained by averaging the Huronian sediments (secondary magnetization) and Nipissing diabase (primary magnetization) site means (Fig. 3, Table 7) is probably a good estimate ($\theta_{63} = 11°$) of the direction of the field at time of cooling of the Nipissing diabase (age = 2150 Ma—Table 2).

Grouping D. This magnetization has high RCF (>1000 G) and blocking temperatures of 500 to 600 °C. It is slightly less stable and probably younger than the C magnetization. A reversal may have occurred during its acquisition at some sites.

Grouping E. Thermal and AF treatments indicate blocking temperatures and MDF comparable to those of magnetization D. However,

TABLE 8. Corroborative results

Index	N	Direction $D°,I°$	Pole position Lat. °	Long. °E	Rock name	Reference
C1	18	259, +82	42N	258	Nipissing diabase (Cobalt)	This paper
C2	07	297, +81	53N	254	Nipissing diabase	Patel 1971
C3	04	042, −74	23N	259	Cobalt Group (contact)	Symons 1967
C4	08	282, +75	45N	237	Nipissing diabase (Elliot Lake)	Symons 1971
C5	14	269, +65	32N	228	Abitibi dikes swarm	Fahrig *et al.* 1965
C6	22	264, +63	28N	227	Abitibi dikes swarm	Larochelle 1966
C7	9	339, +73	73N	238	Nipissing diabase (Gowganda)	Symons & Londry 1975
D1	03	017, −44	15S	264	Nipissing diabase (Cobalt)	This paper
D2	11	009, −40	20S	272	Nipissing diabase (Cobalt)	Symons 1970
D3	04	358, −43	19S	279	Nipissing diabase (Elliot Lake)	Symons 1971
D4	04**	007, −41	19S	274	Nipissing diabase (Cobalt)	Patel 1971
D5	04**	356, −52	11S	280	Nipissing diabase (Matienda)	Patel 1971
D6	11	020, −42	15S	261	Nipissing diabase (Gowganda)	Symons & Londry 1975
D7	03	353, −35	22S	288	Abitibi dikes swarm	Larochelle 1966
E1	09	271, −26	09N	001	Nipissing diabase & Cobalt Group	This paper
E2	11	266, +28	08S	023	Nipissing diabase	Patel 1971
E3	01	084, +33	18N	337	Nonacho seds. (site 001)	McGlynn *et al.* 1974
E4	02	252, +30	01N	031	Nipissing diabase (Elliot Lake)	Symons 1971
E5	39	024, −15	16N	035	Stark Formation	Bingham & Evans 1975

Composite mean index‡	N*	Direction $D°,I°$	Pole position Lat. °,	Long. °E	κ	Composite mean Site unit weight	Work unit weight
CA	72	266, +73	37N	240	31	C1−C6	
CB	6*	263, +75	38N	244	61		C1−C6
CC	37	267, +81	43N	255	27	C1−C4	
CD	4*	260, +80	41N	254	104		C1−C4
CA[1]	81	272, +73	40N	240	27	CA+C7	
CB[1]	7*	274, +75	42N	244	46		CB+C7
CD[1]	46	285, +80	48N	251	26	CC+C7	
CC[1]	5*	281, +80	47N	250	61		CD+C7
CB[2]	7*	263, +75	37N	243	70		CB+OT
CB[3]	8*	271, +76	42N	243	52		CB[1]+OT
DA	37	010, −43	17S	271	23	D1−D6	
DB	6*	008, −44	16S	273	101		D1−D6
DA[1]	40	008, −43	17S	272	23	D1−D7	
DB[1]	7*	006, −43	18S	274	77		D1−D7

N = number of sites; N* = number of works.
**D4 and D5 are based on 19 and 18 specimens, respectively; N has been reduced to 4 in this table so as not to unbalance the weighting with respect to site means of the other works in the DA averaging.
‡Index for composite mean obtained by averaging the first part data in the manner described in the last two columns.
OT = Otish Gabbro (pole: 253 °E, 35 °N).

there is evidence that magnetization E is younger than magnetization D, and that it was acquired during faulting.

Evidence for the three (C, D, and E) magnetizations can be found in all previous studies of the Nipissing diabase. This is listed in Addendum A. The Patel and Palmer (1974) sites are not listed specifically but are included in Patel's work (1971). All sites listed were AF cleaned. In some instances, it was difficult to give the proper assignment because of the information available, and undoubtedly some of the sites may have been wrongly assigned or deleted. Hence, sites 11A and 34A (Patel 1971; Patel and Palmer 1974) appear to belong to Grouping C, but were deleted to comply with the authors' decision. However, the fact that at least three magnetizations can be detected in so many works by different workers leaves little doubt that the magnetizations can be attributed to major tectonic events. This is all the more evident when other igneous bodies are considered. The results from the Abitibi dikes of comparable age (Table 2) and from the same general area are also listed in Addendum A. It is noted that the predominant magnetization in the Abitibi dikes belongs to the C Grouping and that the age determination of the dikes with which the dif-

ferent groupings are associated are not significantly different; for example, the ENEN dikes appear to yield the older age, but then the limits of error are very large (± 279 Ma). Therefore, no correlation between dike orientation and age can be made, based on the age determination, and the grouping assignment has been made according to the different directions, which presumably indicate different ages of magnetization and not necessarily different ages of injection.

The mean direction and pole position obtained according to the assignment given in Addendum A are listed in Table 8. Composite means (*e.g.* CA, ...) calculated using site or work as unit weight and including or excluding the Abitibi dikes are given. Because pole C7 lies significantly to the north of other C poles, composite means including or excluding C7 are also given. Means (CB^2 and CB^3) including the Otish Gabbro discussed later are also given.

Discussion

A polar path for the interval -2600 to -1500 Ma has been drawn in Fig. 11. This path is consistent with all the results available from paleomagnetic studies (see Addendum B). Many uncertainties still exist, and dashed lines and the width of the path (20° in space) have been drawn to emphasize this point. Therefore, owing to the lack of data between -2600 and -2300 Ma, the portion of the path prior to 2300 Ma is not included in the following discussion.

Two features are clearly shown: a latitudinal change followed by an excursion to the east. The apparent polar motion from north to south is similar to Track 5, proposed by Irving and Lapointe (1975) and Pullaiah and Irving (1975), except that the path has been extended from -2200 to -2300 Ma and it is shown that the Nipissing diabase is not at odds with a southwards apparent polar motion, but is in accordance and supports the sense of the motion with the addition of two well defined poles. The placing of the Nipissing C (primary) pole at $\simeq 258°$ E, 42° N is well substantiated from this and other works (C1–C4). The D magnetization yields a pole to the south (D1–D6) at the -1900 to -1800 Ma portion of the path and represents a secondary magnetization, probably acquired at the beginning of the Hudsonian orogeny. That these two (C and D) magnetizations were acquired during major tectonic events is enhanced by the fact that the poles (C5, C6, and D7) obtained from the Abitibi dikes are in such good agreement with their Nipissing counterparts.

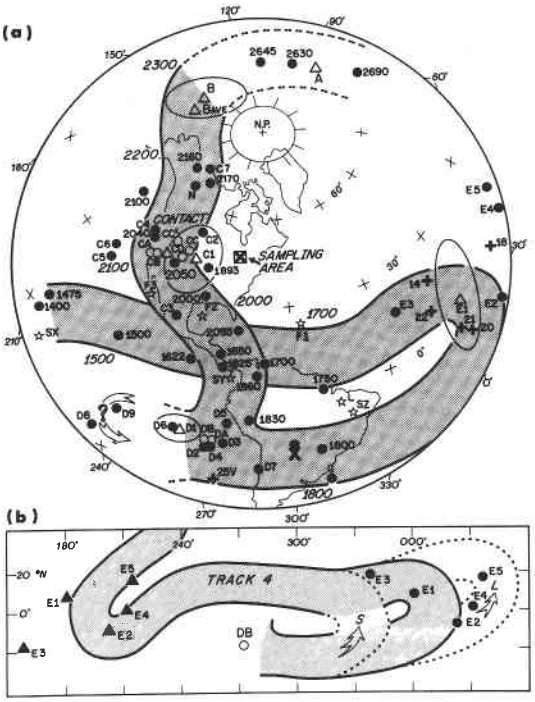


FIG. 11. A path of apparent polar wander relative to Laurentia for the interval 2600–1500 Ma ago. The 20° width serves to emphasize the uncertainties associated with the determination of paleopoles of that age, and the dashed lines serve to draw attention to the fact that no paleomagnetic result is available for an interval of more than 300 Ma ($\simeq 2650$ to 2300 Ma). *Triangles*: This work: A, B, C1, D1, and E1 for the five groupings given with ellipse of confidence ($P = 0.05$) except for A, which is based on a single site; B ave is a rough estimate of the average pole position during the entire magnetization process of Grouping B magnetization; B ave postdates B, which is the pole position during the early phases of magnetization; the pole derived from the contact test is shown. *Crosses*: This work: poles obtained from individual site means of Grouping E and 25 V from vector analysis. *Dots*: Other C, D, and E (numbered) are poles obtained from other studies (Table 8 and Addendum A). *Circles*: Averages of previous poles as listed in Table 8. *Dots*: Numbers 2690 to 1400 associated with poles from other studies are the age estimates in Ma (Addendum B). *Asterisks*: Poles obtained from secondary magnetizations in other works (Addendum B). Poles X and G are discussed in the text. Perspective projection. Other possible apparent polar paths are shown below (b) (mercator projection).

The correlation that can be made between the Otish Gabbro and the Nipissing diabase is noteworthy. The Otish Gabbro pole (Fahrig and Chown 1973; 2050 pole in Fig. 11) falls in the

cluster of the Nipissing C poles. The Otish Gabbro intrudes the Otish Mountains Group and the Nipissing intrudes the Cobalt Group. It has been suggested (Frarey and Roscoe 1970) that the Otish Mountains Group and the Cobalt Group may be time-equivalent. The agreement among poles of the Nipissing diabase, the Abitibi dikes, and the Otish Gabbro (Table 8; CB^2 and CB^3 with $\kappa > 50$) suggests that these igneous bodies were emplaced at approximately the same time into rocks of perhaps the same age. Agreement between age determination of the Nipissing diabase and the Abitibi dikes (Table 2) supports this view. The age of the Otish Gabbro is not accurately determined, but appears to be between −2000 and −2200 Ma (Fahrig and Chown 1973). Therefore, both the paleomagnetic and age determination results indicate that the Nipissing diabase, the Abitibi dikes, and the Otish Gabbro were slightly different phases of widespread igneous events that took place at about −2150 Ma. The data (Fig. 11 and Addendum B) show that several igneous events took place within a few tens of millions of years preceding and following that date; that is, between approximately −2200 to −2100 Ma. Since that period immediately follows the deposition of Huronian sediments, we suggest the name *Post-Huronian Igneous Events* to identify as a group all igneous events that occurred during that time range. Accordingly, specific events as, for example, the Big Spruce complex (−2170 Ma) and the Indin dikes (−2093 Ma) are described as early and late phase events, respectively.[3]

The N pole (mentioned earlier) falls on the polar path very close to two other poles 2160 and 2170 Ma old, indicating the possibility of a remagnetization at about that time. It also lies midway between the B and the CONTACT poles obtained (respectively) from the initial and secondary magnetizations found in the Firstbrook red beds. This may indicate that the N magnetization is the resultant of two unresolved magnetizations, one of which may be initial but superimposed by a secondary magnetization younger than 2200 Ma (perhaps caused by the intrusion of the Nipissing diabase). Owing to these uncertainties and lack of information, a magnetization age has not been assigned.

The southward latitudinal change (Track 5), which is well substantiated by other results shown in Fig. 11 and listed in Addendum B, terminates with the well defined D grouping. The apparent polar path then moves into a longitudinal displacement (Track 4; Irving and Lapointe 1975). A westward excursion may first have taken place. This possibility is owing to the position of D8 and D9; however, those poles are based on 5 and 2 sites, respectively, and may be the result of undifferentiated mixed D magnetizations. For example, the dispersion of D8 is greater than those of the other (C6 and D7—Table 8) groupings (see Fig. 5 of Larochelle 1966), and it is possible that these sites were magnetized during the reversal that occurred at about that time. This would cause a departure from the D grouping as evidenced by the pole (25V) obtained by vector analysis from our site 25. Therefore, the occurrence of a westward excursion at that time is far from being established.

The many results obtained lately by different workers indicate that an excursion towards the east took place between 1850 and 1600 Ma ago. The apex appears to be roughly defined by the E poles listed in Table 8 and shown in Fig. 11. These poles are obtained from secondary magnetizations (which accounts for the dispersion) except for the Stark Formation (E5), which appears to have been magnetized between −1650 and −1800 Ma (Bingham and Evans 1975). Our E1 magnetization has been associated with the faulting that may well have occurred towards the end of the Hudsonian orogeny. Therefore, the apex is given a tentative date of about −1750 Ma. The results from the Flin Flon B (Park 1975) at 1800 Ma and the Et-Then group (Irving *et al.* 1972) at 1750 Ma support both the excursion and its age. This path is also defined

[3]Very recently published results of the Molson dikes (Ermanovics and Fahrig 1975) yield a pole at 246° E, 28° N, which (if plotted on Fig. 11) falls on the polar path and only 7° south of the Otish Gabbro. According to our calibration of the path, this yields an age between −2100 and −2000 Ma, which agrees well with an age of −2000 to −1800 Ma as estimated by Ermanovics and Fahrig. Assuming (since there is no thermal work or contact test) that the magnetization is primary, as argued by the authors, then the intrusion is a late phase of the 'Post-Huronian Igneous Events'.

A second direction obtained from *one site* yields a pole at 296 °E, 05 °S, which falls between the southern and northern legs of the eastward excursion (discussed later) and just outside the descending path. The age of this magnetization could be about −1800 or −1700 Ma, depending on the leg to which the pole belongs. If it belongs to the southern leg, it is near the D Grouping poles and the magnetization was probably acquired at the beginning of the Hudsonian orogeny.

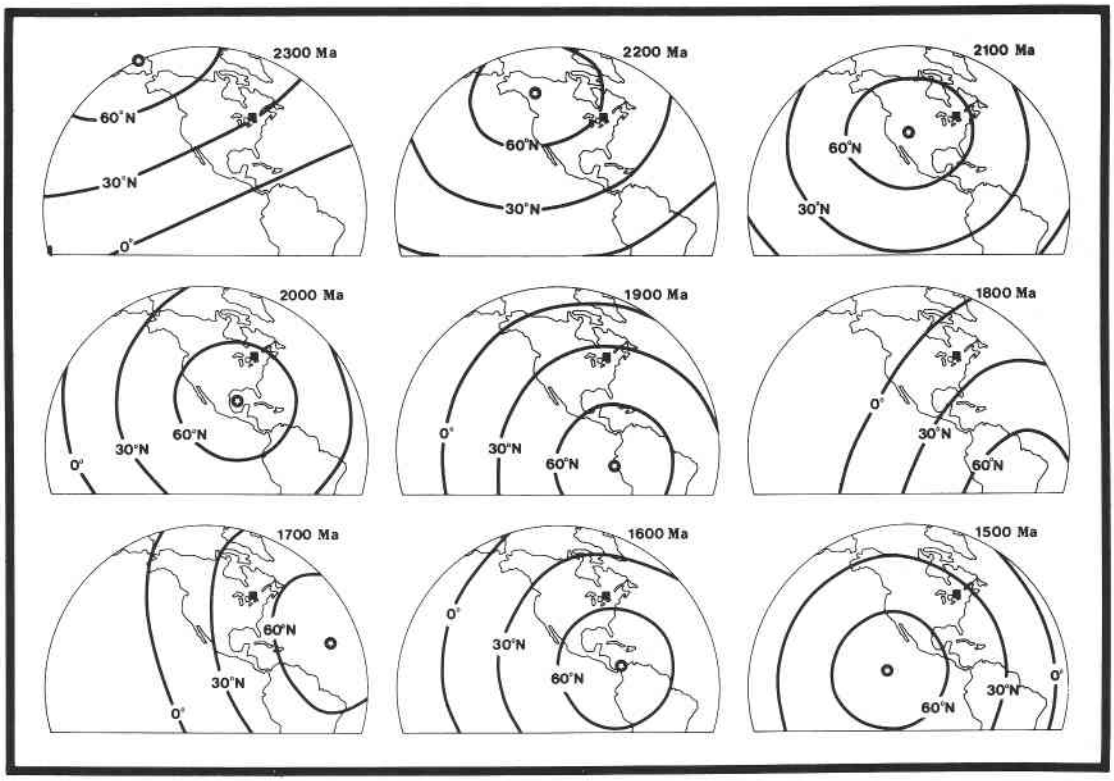


FIG. 12. Latitude maps of Laurentia for the interval −2300 to −1500 Ma, using the polar path of Fig. 11 as the basic datum. The latitudinal motion of Laurentia from 2300–1800 Ma ago is clearly shown; its location was at 2100 Ma near the pole, and at 1800 Ma near the equator. From 1700–1500 Ma ago, Laurentia remained in intermediate latitudes (30°–60° N).

by the relative pole positions of the Kahochella Formation (1860), the Stark Formation, and the Et-Then Group (1750), which stratigraphically lie one above the other with the Kahochella being the lowest. Other poles (F1, L, and SZ) obtained from secondary magnetizations can be explained by this eastward excursion. It is noted that the Dogrib Dikes (G) and the X dikes (X) poles fall on the polar path at about −1800 Ma, although the age of the rock is probably much older; an age of −2692 ± 80 Ma for the Dogrib dikes has been obtained by Rb/Sr method (Gates and Hurley 1973). According to McGlynn and Irving (1974), there is no direct evidence that "the stable magnetization in the dikes was acquired at the time of cooling following intrusion", so that the possibility of a remagnetization at about the time of the Hudsonian orogeny cannot be disregarded. However, a two-stage demagnetization in AF of 400 G followed by thermal cleaning up to 600 °C shows no appreciable change in direction, suggesting that the magnetization might be initial; if so, the reversal of the G pole (130° E, 35° N) places it at about 30° to the south of the 2630, 2645, and 2690 poles.

This longitudinal path may indicate a pole movement relative to Laurentia or it may reflect a dextrorsum rotation of Laurentia (−1850 to −1750 Ma ago) followed a sinistrorsum rotation from −1750 to −1600 and perhaps −1500 Ma. This would explain the often-found secondary magnetizations yielding a streaking of poles along the present-day equator (*e.g.* McGlynn *et al.* 1974). The tracking of the apparent polar path slightly north of the equator, for the interval −1750 to −1500 Ma, is defined by a number of poles (1750, 1700, 1650, 1622, and 1500; Addendum B) falling progressively to the west. This placing of the path between 0° and 30° N latitude is also substantiated by overprints (hard secondary magnetizations) observed in

Flin Flon greenstones (poles F1, F2, and F3; Park 1975) and the Big Spruce complex (poles SY and SX; Irving and McGlynn 1976*a*). For a more detailed discussion of this part of the path (track 4), one may be referred to Irving and McGlynn (1976*b*).

The path is also shown in the inset (different projection) with two options for a short (S) and a long (L) eastward excursion. To satisfy the S path, all E poles are relocated at their antipoles and it must be argued that the E magnetizations were acquired at about −1400 Ma. This option raises several difficulties. The magnetization of the Stark formation (E5) is most probably older than 1650 Ma, owing to the presence of a field reversal. The magnetization yielding the E3 pole has blocking temperatures higher than those of the magnetizations yielding poles between −1700 and −1400 Ma on track 4 (McGlynn and Irving 1975), indicating that it is probably older and to the east. The L path comprises the Stark Formation (E5) pole. If the path represents a linear displacement of Laurentia relative to the pole, this means that from −1800 to −1700 Ma, the displacement averages 12 cm/y. Alternatively, the apparent polar path could be produced by rotation of Laurentia about 260° E, 60° N; this requires an average angular rotation of less than 2°/Ma.

Using the polar path of Fig. 11 as the basic datum, latitude maps of Laurentia have been drawn for the period −2300 to −1500 Ma (Fig. 12). From 1500 to 1900 Ma, Laurentia remained in intermediate to low latitudes. The eastward excursion is well shown by the pole shift on the 1700 and 1800 Ma maps inserted between the look-alike 1600 and 1900 Ma maps. From 2000 to 2200 Ma, Laurentia remained in high latitudes. The 2300 Ma map indicates that Laurentia was then in intermediate latitudes. This map is based on the result of Grouping B, which indicates that at the time of magnetization the sampling area was at latitude 34°. Since this magnetization is of chemical origin, the time of magnetization cannot be established with great accuracy and calibration of the polar path may not be exact. However, the contact tests show that the secondary (TRM) magnetization was acquired at about −2150 Ma. So, the initial (chemical) magnetization B is certainly older than 2150 Ma and probably appreciably older than 2200 Ma according to the scaling of the relative position of the B, Otto Stock (2160 Ma), and Big Spruce Complex (2170 Ma) poles. Therefore, the calibration is probably not in error by more than a few tens of Ma.

Although the possibility that the magnetization may have been acquired after an ice retreat cannot be entirely discounted, the fact that the B pole yields a relatively low latitude, that it was derived from the early phases of magnetization, and that little polar motion occurred at that time, favours a non-glacial origin for the Firstbrook Member. Furthermore, no field evidence (such as dropstones) for a glacial origin could be found, and as suggested by Frarey and Roscoe (1970), "the material in the [Gowganda] formation could be accounted for by non-glacial sedimentary processes". Whether Laurentia was in high latitudes prior to Upper Gowganda time cannot be determined paleomagnetically, since no paleomagnetic record is available for the 350–400 Ma interval separating the B pole and the 2630, 2645, and 2690 poles (Fig. 11). A yet undetected polar excursion could have taken place during that interval.

Acknowledgments

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Addendum A

Following is a list of the sites used (from other works) in the calculation of the mean of our different groupings.

Author	Rock name	Sites used	Grouping
Fahrig *et al.* 1965	Abitibi dikes	All sites	C5
Larochelle 1966	Abitibi dikes	1–7, 9, 11, 12, 16, 18, 19, 20, 22, 24–30 (ENEW)	C6
		10, 21, 23 (ENEN)	D7
		8, 13, 14, 15, 17 (NNE)	D8
Symons 1967	Nipissing diabase and Cobalt Group	Nipissing diabase and upper slate (2 sites)	D9
		Cobalt Group excluding the upper slate	C3
Symons 1970	Nipissing diabase, Cobalt area	All sites	D2
Symons 1971	Nipissing diabase, Elliot Lake area	1, 2, 4, 10, 11, 13, 17, 19	C4
		14, 15, 16, 18	D3
		5, 6	E4
		7, 8, 9	R*
		3, 12	R**
Patel 1971	Nipissing diabase, Cobalt area	ND-2, 3, 4, 5A, 5B, 6A, 6B, 7A, 8, 9, 10, 13, 14, 15, 16, 27, 29, 30, 35	D4
		12, 21, 31	C2
		17, 18, 23, 25	E2
		1, 19, 22, 26	R*
Patel 1971	Nipissing diabase, Lake Matienda area	ML-1, 4, 5, 6, 8, 11, 13, 16, 17, 18, 19, 20, 31; 2A, 3A, 6A, 8A, 36A	D5
		10, 37A	C2
		22, 35A	E2
		15, 21, 25, 29; 9A, 10A, 11A, 34A	R*
		14, 23, 30	R**
Patel 1971	Nipissing diabase, Whiskey Lake area	WL-2, 10	C2
		1, 14, 20, 23, 28	E2
		5, 6, 8, 13, 15, 16, 21, 22, 24	R*
		11, 12, 17, 18, 27, 29, 31	R**
Bingham and Evans 1975	Stark Formation	All sites	E5
Symons and Londry 1975	Nipissing diabase Gowganda	12, 13, 18, 19, 21, 22, 24, 25, 27	C7
		1, 2, 3, 5, 6, 7, 8, 9, 10, 29, 30	D6
		4, 11, 14, 15, 16, 17, 20, 23, 26, 28	R*

R*—Rejected by the author(s).
R**—Not used because of apparent instability of magnetization and/or uncertainty as to which group they belong.

Addendum B

Check-list of paleomagnetic poles

The rock units from which the poles were obtained are listed in order of increasing estimated age (see Irving and Lapointe 1975 for age assignments). It must be emphasized that these age estimates are approximate only, and are merely the best we can make of a woefully incomplete record. The errors associated with these age estimates vary from 10^7 to 10^8 years, and the final digit is not significant—these are given here in order to identify each result. Further details may be obtained from the first (Hicken *et al.* 1972) and second (Irving and Hastie 1975) issues of the Ottawa catalogue, where information about age determinations and qualitative statements about age estimates are given.

Age	Name	Ottawa list number
1400	Michikamau Anorthosite	205
1475	Croker Island Complex	129
1500	Western Channel diabase	504
1622	Melville Daly Bay Metamorphics	456
1650	Nonacho Group	411
1700	Sparrow dikes	410
1750	Et-Then Group	341
E	Group E, present work	—
1800	Flin Flon B magnetization (Park 1975)	573
D	Group D, present work	—
1825	Dubawnt Group	481
1830	Martin Formation	468
1860	Kahochella Formation	—
1893	Spanish River Complex	413
2000	Gunflint Formation	191
2040	Wind River Dikes	326
2050	Otish Gabbro	467
2093	Indin dikes	492
2100	Mugford basalt	314
C	Group C present work	—
N	Gowganda FM Elliot Lake area (Symons 1975)	546
2160	Otto Stock (Pullaiah and Irving 1975)	543
2170	Big Spruce Complex (Irving and McGlynn 1976*a*)	555
B	Group B, present work	554
A	Group A, present work	—
2630	Chibougamau Greenstone Belt (Ueno and Irving 1976)	581
2645	Stillwater Complex	244
2690	Matachewan dikes	107

Other poles plotted in Fig. 11

F_1, F_2, F_3	Secondary	(Park 1975)
Sx, Sy, Sz	Secondary	(Irving and McGlynn 1976*a*)
L, Duck Lake	Secondary	(McGlynn and Irving 1975)
G, Dogrib dikes		(McGlynn and Irving 1975)
X, X-dikes		(McGlynn and Irving 1975)

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